

Electrostatics Formula Sheet

Coulomb's Law: $F = k * q_1 * q_2 / r^2$ ($k = 9e9 \text{ N m}^2 / \text{C}^2$)

Electric Field: $E = F / q = k * Q / r^2$

Electric Potential: $V = k * Q / r$

Potential Energy: $U = k * q_1 * q_2 / r$

Capacitance: $C = Q / V$

Parallel Plate: $C = \epsilon_0 * A / d$

Energy in Capacitor: $U = 1/2 * C * V^2$

Gauss's Law: $\text{flux} = Q_{\text{enclosed}} / \epsilon_0$